

Direct DC Coupling of a PEM Electrolyser to a Wind-Turbine Generator

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This 2-pager assesses the feasibility of directly coupling a wind-turbine generator to a PEM electrolyser through a shared DC system, without a dedicated electrolyser-side DC/DC converter. The focus is on the underlying operating principle, the conditions required for safe and stable operation, and the extent to which existing research and industry projects support the concept.

1. Objective and conclusion

Wind-turbine generator → active rectifier/up-chopper → DC link → PEM stack or stack array

The proposed system has no AC export connection, grid-side inverter, electrolyser rectifier or separate stack-current-regulating DC/DC converter. The active rectifier/up-chopper remains part of the wind-turbine generator and controls generator current, electromagnetic torque and therefore the power extracted by the turbine.

The PEM stack is connected to the resulting DC link. Its polarization characteristic defines the available voltage–current combinations, and the operating point moves continuously as generator power changes:

$$P_{\text{gen}} = V_{\text{stack}}(I_{\text{stack}}) \cdot I(I_{\text{stack}})$$

The turbine controls the supplied power; the stack characteristic determines the corresponding DC voltage and current. The DC-link voltage is therefore an operating result rather than the primary controlled variable.

Conclusion. A dedicated electrolyser-side DC/DC converter is not inherently required. Its normal functions can be allocated to generator power control, generator–stack voltage matching, supervisory operating limits and coordinated protection.

2. Operating principle

Power input is controlled by the generator. As generator power rises or falls, the PEM operating point moves along its polarisation curve to a new voltage and current.

There is no need for an active-front-end voltage-control loop during normal hydrogen production. DC voltage is still measured for converter limits, operating-envelope supervision and protection, but it is not held at one fixed reference. The architecture is therefore best described as generator-power-controlled direct DC coupling.

3. Allocation of functions without a separate stack converter

Function	How it is provided
Power and current	Generator torque/current control, limited to the power that the active PEM configuration can safely accept.
Voltage adaptation	Co-design of generator winding voltage, speed range, rectifier/up-chopper range and PEM cell count.
PEM limits	A supervisory permitted-power envelope based on current density, temperature, pressure, gas purity, water and cooling conditions.
Protection and isolation	Coordinated turbine and electrolyser protection. Applicable earthing, insulation and galvanic-isolation requirements remain system design inputs.

4. Feasibility conditions

Most component-level conditions are already standard practice in separate wind turbines and PEM electrolyzers. The new work lies mainly in electrical matching, exchange of limits, timing and allocation of authority across the shared interface.

Electrical matching

The full PEM polarisation envelope, including temperature, pressure, tolerance and ageing, must remain within the turbine voltage and current ratings.

Coordinated control

The turbine receives a continuously updated permitted-power limit. The final command is bounded by available wind power, generator capability and PEM process limits.

Start, stop and faults

Existing turbine functions (current reduction, pitch, braking and isolation) must be coordinated with standard PEM permissives, stack isolation, purge and safe shutdown.

Boost-only constraint

If the power stage can only raise voltage, the **passive** rectified generator voltage must remain below the required stack voltage across the permitted speed range.

Modular stack option

A modular array could be beneficial for low-load operation, availability or maintenance. Branch switching provides coarse capacity selection; generator torque remains the continuous power actuator.

Harmonics and drivetrain

Diode rectification can introduce generator-current harmonics and torque ripple. Generator electromagnetic design and drivetrain loads require validation.

5. Evidence base

The references support the central proposition that a dedicated DC/DC converter is not a fundamental requirement when source and stack are deliberately matched and power is controlled elsewhere in the system.

Direct proof of principle.

Clarke et al. operated a 2.4 kW PV array directly coupled to a PEM electrolyser without MPPT or a DC/DC converter between them. Series–parallel configurations were selected to match the voltage–current characteristics, and the system completed roughly 600 hours of operation. This demonstrates that electrical matching and the PEM polarisation characteristic can establish the operating point without an intervening converter [1].

Wind-specific proof of principle.

RMIT researchers also investigated direct coupling of a fixed-pitch, variable-speed aerogenerator to PEM electrolyzers for remote-area supply. The scale is not representative of a modern multi-megawatt turbine, but the work directly supports matching a wind generator and PEM load without intervening voltage-conversion or MPPT electronics [2].

Industrial architecture support.

ITM Power and Ørsted proposed an electrolyser at an individual Type IV wind turbine, directly electrically connected to the turbine DC link. They identify fewer power-conversion steps, lower losses and removal of AC rectification as benefits. The publication also states that the concept had not yet been demonstrated, so it supports the intended architecture rather than proving this exact converter arrangement at scale [3].

Adjacent electrochemical precedent.

Fraunhofer ICT's RedoxWind project connects a redox-flow battery directly to the intermediate DC circuit of a 2 MW wind turbine, explicitly eliminating an additional conversion step. A battery is not identical to a PEM load, but the project validates turbine-level electrochemical integration and shared DC-side design [4].

Engineering qualification.

Recent DC wind-turbine research confirms the benefit of fewer conversion stages, while generally retaining high-power DC/DC conversion for voltage adaptation, isolation and fault handling. This shows that a DC/DC converter can be useful; it does not establish that one is inherently required to determine the PEM operating point. The same work highlights current harmonics and torque ripple as validation topics for diode-rectifier concepts [5].

6. Validation priorities

Before commercial deployment, validation should focus on the integration-specific questions rather than repeating component qualification already covered by normal turbine and PEM standards.

Operating envelope Verify the matched generator and PEM voltage–current range over speed, temperature, pressure, tolerance and ageing.	Control coordination Demonstrate generator-power control against the PEM polarisation model, including low-load transitions and branch switching where used.
Fault and sequencing Validate start permission, load rejection, communication failure, isolation and safe shutdown with agreed timing and authority.	Electrical and mechanical effects Measure rectifier harmonics, torque ripple, thermal loading and current sharing before progressing to a multi-megawatt demonstrator.

7. Final assessment

Direct DC coupling of a wind-turbine generator and PEM electrolyser without a dedicated electrolyser-side DC/DC converter is technically credible. The turbine controls generator power, while the PEM polarisation characteristic establishes the corresponding DC voltage and current.

Many required functions are already standard within independent wind turbines and electrolyzers. The genuinely integration-specific tasks are generator–stack voltage matching, real-time exchange of the PEM power-acceptance envelope, coordinated sequencing and protection, and validation of harmonics and drivetrain effects.

A separate electrolyser-side DC/DC converter is not a fundamental requirement. The remaining gap is full-scale engineering validation—not proof of the underlying electrical principle.

Key references

- [1] R.E. Clarke et al., "Direct coupling of an electrolyser to a solar PV system for generating hydrogen", International Journal of Hydrogen Energy 34 (2009) 2531–2542. <https://doi.org/10.1016/j.ijhydene.2009.01.053>
- [2] A. Chantharasenawong et al., "Direct coupling of a fixed-pitch variable-speed aerogenerator to PEM electrolyzers in a wind-hydrogen system for remote area power supply", 17th World Hydrogen Energy Conference (2008). https://research-repository.rmit.edu.au/articles/conference_contribution/Direct_coupling_of_a_fixed-pitch_variable_speed_aerogenerator_to_PEM_electrolyzers_in_a_wind-hydrogen_system_for_remote_area_power_supply/27369558
- [3] ITM Power and Ørsted, "Wind Turbine Electrolyser Integration", 6 April 2020. <https://itm-power.com/news/wind-turbine-electrolyser-integration>
- [4] Fraunhofer ICT, "RedoxWind – Redox flow battery meets wind turbine". <https://www.ict.fraunhofer.de/en/research/ae/redoxwind.html>
- [5] F. Andresen et al., "Evaluation of DC wind turbine concepts for coupling wind energy with electrolysis", Engineering Research 89, 35 (2025). <https://doi.org/10.1007/s10010-025-00796-0>